



DRAFT · CLINICAL REVIEW COPY

ROOT CAUSE UNLOCKED

The Complete Patient's Guide to *Achilles* Tendinopathy

Why rest does not fix a tendon, what loading actually does, and how to run the program the trials actually ran

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Complete Patient's Guide to Achilles Tendinopathy

Why rest does not fix a tendon, what loading actually does, and how to run the program the trials actually ran

DRAFT · CLINICAL REVIEW COPY

This edition is pending Dr. Seay's clinical sign-off. Every citation has been verified against its live PubMed record, including a check for retractions. The companion citation ledger lists each source and the claims that were investigated but left uncited.

BEFORE YOU START

Contents

What this guide is for

Chapter 1. What is actually wrong in there

Chapter 2. The anatomy that matters to you

Chapter 3. How it started, and what raises your risk

Chapter 4. Measuring yourself properly

Chapter 5. What the timeline actually looks like

Chapter 6. Midportion or insertional, and why it changes your exercise

Chapter 7. The loading evidence, which is the treatment

Chapter 8. The in-clinic tissue prep protocol

Chapter 9. Everything else you will be offered, assessed honestly

Chapter 10. Nutrition, and the one trial that was actually run in your tendon

Chapter 11. Looking above and below the tendon

Chapter 12. Red flags, which are not negotiable

Chapter 13. Building your recovery plan

Final thoughts

References

Working with Root Health

What this guide is for

The Achilles is the strongest tendon in your body and the one most famous for complaining. If you have the stiff, aching, sometimes thickened tendon that hates the first steps of the morning and the first mile of every run, you have landed on one of the clearest treatment stories in all of musculoskeletal care.

It is also the opposite of what almost everyone does first.

Rest calms an angry tendon. It does not rebuild it. Progressive loading rebuilds it. That sentence is the whole guide, and the rest of these pages exist because “just load it” is useless advice without knowing how much, how often, for how long, which version for your particular tendon, and how to tell the difference between productive discomfort and a dose that outran you.

I have written this to be more useful than what you will find in an afternoon of searching, and there are three specific places where it is: the distinction between midportion and insertional disease, which changes the exercise you should be doing and which most sources gloss over; a self-scoring questionnaire so you can measure yourself rather than guess; and the actual in-clinic protocol I use with these patients, tools and timings included.

This guide is educational. It does not diagnose your tendon and it does not replace an examination. What it will do is let you have a far better conversation with whoever examines it.

Chapter 1. What is actually wrong in there

Start by discarding the word most people bring with them: inflammation.

Tendinopathy is not primarily inflammatory in the classic sense. That is precisely why anti-inflammatory strategies keep disappointing people who try them, and why the guy at the gym who took ibuprofen for six weeks is still limping. What is happening is better described as failed adaptation. A tendon was asked to absorb more load than it had been prepared for, and instead of adapting cleanly it responded with disorganised collagen, thickening, increased fluid, new blood vessel and nerve ingrowth, and pain.

That last item matters more than it sounds. The pain in tendinopathy does not track neatly with the damage on a scan. Some very ugly-looking tendons are pain free, and some tendons that image reasonably well hurt considerably. This is why I will not chase your scan, and why the measures in Chapter 4 are about what you can do and how you feel rather than what a picture shows.

The good news buried inside all of this: tendon is living, adaptive tissue. Load it correctly and it remodels. In one of the eccentric training studies, tendon thickness decreased and structure normalised at follow-up alongside the clinical improvement (PMID 14751936). You are not managing a permanent injury. You are re-training a tissue that got behind on its homework.

Chapter 2. The anatomy that matters to you

You do not need a full anatomy lesson. You need four things.

The tendon itself. The Achilles is the combined tendon of the gastrocnemius and soleus, running from the calf down to the back of the heel bone. It is the strongest tendon in the body and routinely handles many times your body weight during running and jumping.

Two calf muscles, not one. The gastrocnemius crosses both the knee and the ankle. The soleus crosses only the ankle. This is not trivia: it is why your rehab has to include both a straight-knee and a bent-knee version of the same exercise. Straight-knee work biases the gastrocnemius, bent-knee work biases the soleus. Do only one and you have trained half the system.

Two different problem locations. Midportion tendinopathy sits roughly two to six centimetres above the heel bone. Insertional tendinopathy sits right at the heel bone where the tendon attaches. They feel similar to the person who has them. They do not respond to the same exercise, and Chapter 6 is entirely about why.

The calf as a spring. During running, the Achilles stores and returns elastic energy. When calf capacity drops, that spring gets stiffer and less forgiving, and the tendon absorbs load it was not built to absorb alone. Which is the real reason calf strength work is the treatment rather than a nice extra.

Chapter 3. How it started, and what raises your risk

Almost every case I see has a load story, and it is usually one of these.

The spike. A sudden increase in running volume, hills, or speed. A new sport. A return after time off, at the intensity you left rather than the intensity you currently have. Or, most commonly, years of quietly declining calf capacity meeting one ambitious month.

The surface or footwear change. New shoes with a different heel-to-toe drop change how much the tendon is asked to do. A drop from a higher heel to a lower one increases Achilles demand, and doing that overnight rather than over weeks is a classic trigger. So is moving training onto hills, or onto sand.

The medication. Fluoroquinolone antibiotics carry a genuine, documented association with Achilles tendon problems including rupture. A population-based cohort study found that within 90 days of first fluoroquinolone use the standardised incidence ratio for Achilles rupture was 3.1, roughly triple the expected rate, though the absolute incidence among users remained low (PMID 17334751). The risk appears higher when a fluoroquinolone is taken alongside oral corticosteroids. If your tendon pain started after a course of one of these antibiotics, tell your prescriber. That is a specific and genuinely useful piece of information, and most people never think to mention it.

The background factors. Increasing age, higher body mass, diabetes, and reduced ankle dorsiflexion range all appear across the risk literature. None of them is destiny, and two of them are modifiable.

I want to be honest about something here. A great deal of what circulates about tendinopathy causes, particularly around foot posture and specific gait faults, is a mixture of plausible mechanism and weak evidence. The load story is the part that has held up. Start there.

Chapter 4. Measuring yourself properly

Most people track their tendon by vibes. That is why they cannot tell whether the last six weeks helped.

There is a validated questionnaire for exactly this condition. The VISA-A was developed and tested by the Victorian Institute of Sport Tendon Study Group and published in 2001. It showed good test-retest reliability at $r = 0.93$ and good stability when repeated a week apart at $r = 0.81$, and it distinguished patients across a range of severity from controls (PMID 11579069). It scores out of 100, where 100 is a fully asymptomatic, fully active tendon.

Score yourself today, write the number down, and score yourself again every four weeks. Ask your clinician for the full form, or work through its domains, which are:

- How stiff is the tendon when you first get up in the morning, and for how many minutes?
- Do you have pain when walking on flat ground?
- Do you have pain when walking up stairs?
- Do you have pain during or immediately after 30 minutes of walking?
- Do you have pain when doing a single-leg heel raise, and how many can you manage?
- Can you hop, and how many single-leg hops before pain?
- Are you currently doing sport or activity, and at what level and with how much pain?

Alongside the VISA-A, track two numbers weekly. Morning stiffness, in minutes until the tendon feels normal. And pain from 0 to 10 during a standard task you repeat every week: the same hill, the same flight of stairs, the same distance.

Those three numbers together, the VISA-A monthly and the two weekly measures, will tell you within six weeks whether your program is working. Nothing else will.

Chapter 5. What the timeline actually looks like

Tendon adapts slowly. This is the single hardest thing about the condition and the reason most people abandon a program that was working.

Weeks 0 to 2. Symptoms often feel slightly worse before better as loading starts. Morning stiffness is the last thing to improve and the least reliable early signal.

Weeks 2 to 6. The first honest signal. Pain during your standard task should be trending down even if it has not gone. In the studies, this is where the curves start to separate.

Weeks 6 to 12. Meaningful change in function. The trials that produced good results ran twelve weeks (PMIDs 9617396, 26018970). If you stop at six because it is not fixed, you stopped in the middle.

Three to twelve months. Continued structural adaptation and return to full activity. The one-year follow-up in the Beyer trial showed the clinical gains held at 52 weeks (PMID 26018970).

If you are not trending better by six weeks of honest loading work, that is not a reason to load harder. That is a reason to be re-assessed, because either the diagnosis, the dose, or the location classification needs revisiting.

Chapter 6. Midportion or insertional, and why it changes your exercise

This is the chapter I most want you to read, because getting it wrong is the most common reason a well-intentioned home program fails.

The classic eccentric protocol involves lowering your heel below the level of a step, taking the ankle into dorsiflexion. For midportion tendinopathy that is the protocol that was studied and it works.

For insertional tendinopathy, where the problem sits at the heel bone itself, that same dorsiflexion compresses the tendon against the bone. You are actively aggravating the tissue you are trying to rehabilitate.

The evidence on this is specific and it is worth quoting. A study of patients with chronic insertional Achilles tendinopathy used a modified eccentric regimen performed on flat ground, without loading into dorsiflexion. In that group 67 percent were satisfied with the result, compared with the roughly 32 percent good outcome previously seen in insertional patients using the dorsiflexion protocol (PMID 18184750). A later review made the same point in its title: eccentric training is not for everyone (PMID 30042626).

So, practically:

- **Midportion** (pain two to six centimetres above the heel bone, often with a thickened tender spot you can pinch): heel raises off a step, lowering below level, is appropriate.
- **Insertional** (pain right at the back of the heel bone, often worse with shoes that press on it, worse going uphill): work on flat ground. Stop at neutral. Do not drop the heel below level, at least in the early phase.

If you cannot tell which you have, that is a genuine reason to be examined rather than to guess. It takes about ninety seconds to determine in person and it changes everything downstream.

Chapter 7. The loading evidence, which is the treatment

Here is the part that actually fixes tendons.

The original protocol. Alfredson's 1998 study took fifteen recreational athletes with chronic Achilles tendinosis who had failed conventional treatment, and ran them through twelve weeks of heavy-load eccentric calf training. All fifteen returned to their pre-injury level with full running activity (PMID 9617396). Small study, dramatic result, and it changed the field. The protocol was three sets of fifteen, twice daily, straight-knee and bent-knee, seven days a week, for twelve weeks. That is 168 sessions, which is exactly the problem.

The modern comparison. In 2015, Beyer and colleagues randomised 58 patients with chronic midportion Achilles tendinopathy to either traditional eccentric training or heavy slow resistance training, three sessions a week, for twelve weeks, with follow-up at 52 weeks. Both produced significant and lasting improvements. There was no meaningful difference in outcome between them (PMID 26018970).

But look at the compliance figures, because this is where the practical answer lives: 92 percent in the heavy slow resistance group versus 78 percent in the eccentric group, and patient satisfaction at twelve weeks tended to favour the heavier, less frequent program.

The conclusion I draw from those two studies together. Both work. Pick the one you will actually do. For most people with a job and a life, that is three heavier sessions a week rather than fourteen lighter ones.

What heavy slow resistance actually means:

- Calf raises, both straight-knee and bent-knee, done slowly. Roughly three seconds up, three seconds down.
- Loaded. Bodyweight alone stops being a stimulus quickly. Add a backpack, a barbell, a machine, whatever you have.
- Three sessions per week, on non-consecutive days.
- Progressed in load as tolerated, working in the range where the last few repetitions are genuinely hard.
- Sustained for twelve weeks minimum before judging it.

And the dose signal from the wider literature points the same direction: greater efficacy with higher intensity and lower frequency, which is the combination that gives a strong stimulus and then allows recovery between sessions.

Do you have to stop running? No, and this is one of the more liberating findings in the field. Silbernagel and colleagues randomised 38 patients with Achilles tendinopathy to either continue tendon-loading activity such as running and jumping using a pain-monitoring model, or to stop those activities for the first six weeks. The group that kept training did as well as the group that rested (PMID 17307888).

The pain-monitoring model, which is the rule I give patients:

- Pain during and after loading, up to about 5 on a 0 to 10 scale, is acceptable.
- It must settle back to your baseline by the following morning.
- It must not climb week over week.
- If any of those three fails, the dose was too high. Reduce it, do not abandon it.

That model is what separates productive rehab from repeated aggravation, and it is why “does it hurt” is the wrong question. The right question is “has it settled by tomorrow, and is it lower than last week.”

Chapter 8. The in-clinic tissue prep protocol

This is my own protocol, and it is the part you will not find anywhere else.

Understand what it is for. None of this rebuilds a tendon. Loading rebuilds a tendon. What this does is reduce the calf and ankle stiffness and tissue tone that make the loading work harder to tolerate and less pleasant to do, particularly in the first few weeks when the tendon is at its most irritable. It is preparation, not treatment. Patients who do it tend to tolerate the strength work better and stick with it, which is the actual mechanism by which it helps you.

Ankle protocol.

- **Foam rolling the calf.** Seated, from below the knee to above the ankle, working the inside and the outside. 60 to 90 seconds per leg, moderate pressure. **Stay off the Achilles itself.**
- **Percussion massager, speeds 1 to 2, level 3 only for intensity.** Tibialis anterior, cushioned head, upper quadrant of the lower leg, 1 to 2 minutes per leg. Ankle ligaments, cushioned head below the ankle bone, inside and outside, 30 to 60 seconds per side per leg, staying off the bone, and skip it entirely if it is too painful. Peroneals, cushioned head or round ball, outer lower leg, 30 to 45 seconds per leg.
- **Firm trigger point tool.** Taut or tender areas around the ankle, inside and outside, 30 to 60 seconds per side per leg.
- **Foam balance pad.** Single-leg balance, 1 to 2 minutes each foot.

Foot protocol, which I add for most Achilles patients because the calf and the plantar surface behave as one system.

- **Percussion device, round or flat tip, level 1 to 2.** Calf upper and lower, 2 to 3 minutes. Tibialis anterior, 1 minute.
- **Massage ball or tennis ball.** Plantar fascia, rolling the bare foot over the ball through the arch and heel and any tender spots, 2 to 3 minutes per foot. Then the heel specifically, seated, small circles with sustained pressure, 60 to 90 seconds per foot.
- **Roller stick.** Calf, from above the Achilles to below the knee, 1 to 2 minutes per leg.
- **Slant board.** Stand at a level that gives a good calf stretch, heel down on the board, 1 to 2 minutes.
- **Firm trigger point tool.** Seated, foot crossed over the opposite knee, working tender areas, rolling back and forth.

Safety, and I mean these.

- Avoid all of this over an acutely injured area in the first 48 to 72 hours.
- Reduce pressure over anything actively inflamed.

- Skip percussion therapy entirely if you have a pacemaker or another implanted electronic device.
- Avoid aggressive pressure over varicose veins or compromised skin.
- Start light. Pain must not exceed 6 out of 10. Back off immediately for sharp or shooting pain.
- **Never work over a calf that is swollen, warm, or tight in one leg only.** See Chapter 12.

Chapter 9. Everything else you will be offered, assessed honestly

Rest. Calms symptoms, changes nothing structurally, and deconditions the calf you are about to need. It is not a treatment. Relative rest, meaning reducing the aggravating load while continuing to load the tendon deliberately, is a different thing and is appropriate.

Stretching. Reasonable for midportion if it feels good. Actively counterproductive for insertional disease, for the compression reason in Chapter 6.

Heel lifts. Genuinely useful short-term for insertional pain, because they reduce dorsiflexion and therefore compression at the insertion. Useful as a symptom tool while you build capacity. Not a fix, and not something to live in permanently.

Corticosteroid injection. I would ask a lot of questions before accepting one into or near this particular tendon. Ask specifically what the loading plan around it is. Ask about rupture risk, especially if you are also on or have recently taken a fluoroquinolone, since the combination shows the highest risk signal in the cohort data. Short-term symptom relief that lets you deload a tendon that needs load is not obviously a win.

Shockwave, injections of various kinds, and the rest of the procedural menu. Some have a reasonable evidence base and some are ahead of it. My position is consistent whichever you are offered: ask what the loading program around it is. A procedure without a rehab plan gives you back the tendon you had.

Anti-inflammatories. Reasonable for short-term symptom management. Not a treatment for the underlying problem, for the reason in Chapter 1.

Surgery. Reserved for genuinely recalcitrant cases after a proper loading program has been tried for months, not weeks. If surgery is being discussed and you have not done twelve honest weeks of progressive loading, that conversation is premature.

Chapter 10. Nutrition, and the one trial that was actually run in your tendon

It would be easy to keep the answer here short: nothing has been properly tested, collagen powders are not worth your money, move on. That would be too blunt on all three counts. The evidence is thin, but it is not absent, and one of the few trials that exists was run in exactly your condition.

Start with what your tendon is made of, because it decides what is worth buying. Roughly three quarters of the dry weight of an Achilles tendon is type I collagen, a rope of protein wound from repeating glycine, proline and lysine. Building it requires your body to chemically modify proline and lysine after the chain is assembled, and the enzymes that do that, prolyl hydroxylase and lysyl hydroxylase, cannot work without vitamin C and iron. The enzyme that cross-links those ropes into something that carries load, lysyl oxidase, requires copper, and manganese and zinc are needed elsewhere on the same assembly line. That is textbook connective tissue biochemistry, not a supplement claim, and it is why supplying protein without supplying cofactors only does half the job.

The timing is the part almost nobody gets right. Collagen synthesis in tendon is driven by mechanical load and rises in the hours after you load the tissue, so the question is not whether to take collagen but whether the raw material is in your bloodstream when your tendon is being told to rebuild. In the study that established this, eight healthy men took vitamin C-enriched gelatin an hour before six minutes of rope skipping, and the 15 gram dose doubled a blood marker of new collagen formation (PMID 27852613). A later dose-response trial found that 30 grams of hydrolyzed collagen with 50 mg of vitamin C an hour before heavy resistance exercise produced more collagen synthesis than 15 grams, while **15 grams was not statistically different from nothing at all** (PMID 38007183). Most products on the shelf sit at or below the dose that failed to beat placebo. If you run this, run it at 30 grams, about an hour before your heel raise session, not at bedtime.

Now the trial in your actual condition, which is easy to miss because it is small and rarely quoted. A three-arm randomized trial enrolled 59 patients with Achilles tendinopathy and compared eccentric training alone, eccentric training plus a supplement of mucopolysaccharides with type I collagen and vitamin C, and passive stretching plus that same supplement. All three groups improved at six and twelve weeks. The supplement added benefit specifically in patients with **reactive, early-stage** tendinopathy, where pain at rest fell further than with eccentric training alone, and it added nothing once the tendon had reached the degenerative stage with established matrix and vascular changes (PMID 28053674). Two of that trial's authors are employed by the ingredient's manufacturer, which is exactly the sponsorship pattern this book flags elsewhere, so weigh it accordingly rather than treating it as settled.

That stage distinction is the most useful thing in this chapter, and it maps onto chapter 1. An early, reactive, angry tendon is still responsive tissue, and that is the window where nutritional support appears to add something. A long-standing degenerative tendon has structural change that a capsule did not reverse in the one trial that looked. So the honest rule is: if your Achilles pain is weeks old, the supplement side is worth

running alongside the loading program. If it has been years, put the money and the attention into the loading program in chapter 7, because that is what has been shown to change the tissue at that stage.

Here is the ceiling, and I would rather you hear it from me than from a refund request. A blood marker of collagen synthesis is not a healed tendon. When an independent group tested whether collagen peptides change tendon structure, giving 15 grams daily through fifteen weeks of resistance training with MRI endpoints, they found no advantage over placebo on tendon size, stiffness or mechanical properties. Both groups improved, and the training did the work (PMID 37436929). The most recent independent systematic review, nineteen studies and 768 participants, found small statistically significant effects and graded the tendon findings specifically as very low certainty (PMID 39060741). So: the mechanism is real, the acute response is real and dose-dependent, one in-region trial is positive in early-stage disease and industry-affiliated, and the one hard structural endpoint tested independently came back null. That is a cheap, low-risk experiment with honest odds. It is not a treatment with proven structural benefit, and anyone selling it to you as one is ahead of the evidence.

What holds regardless of which supplement you run, because it is the fundamentals. Adequate protein supports the tissue you are asking to remodel, and the meta-analysis across 49 studies and 1,863 participants found the benefit plateaus around 1.6 grams per kilogram of body weight per day (PMID 28698222). For a 180 pound person that is roughly 130 grams daily, and most people rehabbing a tendon eat well under it. Collagen is a poor quality protein in isolation, low in the essential amino acids muscle needs, so count a collagen dose as a targeted addition to that total and never as the total itself. Correcting a documented vitamin D deficiency matters on general grounds rather than as a tendon treatment. And addressing metabolic health where diabetes or elevated body mass are in the picture remains one of the highest-value things in this book, because both appear in the risk literature for this condition.

What I would actually do. If your tendon pain is recent and reactive, run 30 grams of hydrolyzed collagen or gelatin with vitamin C an hour before each loading session for twelve weeks, make sure the cofactors are covered, and hit the protein floor. If your tendon pain is long-standing and degenerative, buy none of it and spend the money and the attention on the loading program instead. Either way, set a start date and an end date before you buy anything, change one thing at a time, and let the twelve week mark give you a real answer rather than an open-ended subscription.

Nothing here substitutes for being examined, none of it changes the fact that load is the treatment, and no supplement shortens the months-long timescale on which a tendon actually adapts.

Chapter 11. Looking above and below the tendon

The Achilles does not work alone, and two things regularly show up.

Ankle dorsiflexion range. Reduced ankle mobility appears across the risk literature. If your ankle will not move, your foot and tendon absorb load differently. Worth assessing, and worth working on if restricted.

Hip and single-leg control. How you accept load on one leg determines what arrives at the Achilles. Poor single-leg control means the calf and tendon do more work than they were designed to, every step. This is why single-leg balance sits in the pre-treatment protocol in Chapter 8, and it is why I look at the hip in someone whose Achilles keeps flaring despite good calf work.

I will be measured here: treating the hip does not cure a tendinopathy. But in the patient whose tendon keeps returning after a good loading program, the answer is more often up the chain than in the tendon.

Chapter 12. Red flags, which are not negotiable

A sudden pop or snap at the back of the ankle, often described as feeling kicked, followed by weakness pushing off. That is an Achilles rupture pattern until proven otherwise. Do not stretch it, do not load it, do not massage it. Get evaluated promptly.

Calf swelling, warmth, or tightness in one leg, particularly after travel, surgery, or a period of immobilisation. That is a blood clot concern. Evaluated today, and never massaged or rolled. This is the single most important line in this guide, because the pre-treatment protocol in Chapter 8 involves working on the calf.

A hot, red, swollen area with fever. Infection concern. Same-day care.

Tendon pain that began after starting a fluoroquinolone antibiotic. Prompt conversation with the prescriber (PMID 17334751).

Numbness, pins and needles, or weakness in the foot, which points somewhere other than the tendon.

No improvement by six weeks of honest loading work. Not an emergency, but a genuine reason for reassessment rather than more of the same.

Chapter 13. Building your recovery plan

Weeks 1 to 3. Establish and calm.

- Determine midportion or insertional. Get examined if you are unsure.
- Score your VISA-A. Write it down.
- Start the pre-treatment protocol in Chapter 8, daily or before loading sessions.
- Begin calf loading. Bodyweight or light, both straight-knee and bent-knee. Three sessions a week, non-consecutive days.
- Insertional cases: flat ground only, no dropping below level.
- Reduce, do not eliminate, the aggravating activity. Apply the pain-monitoring model.
- Track morning stiffness in minutes and pain during one standard task.

Weeks 3 to 8. Load and build.

- Add external load, progressively. This is the phase people under-dose, and under-dosing is the most common reason a program fails.
- Keep it slow: roughly three seconds up, three seconds down.
- Maintain three sessions a week. Resist the urge to do it daily.
- Continue the pre-treatment work before sessions if it helps tolerance.
- Re-score VISA-A at week 4. Compare, do not guess.
- Insertional cases may begin cautiously exploring range as symptoms allow, guided by your clinician.

Weeks 8 to 12 and beyond. Strengthen and return.

- Keep progressing load. The tendon has not finished adapting at week 8.
- Introduce faster and elastic work if you are returning to running or sport: hops, skips, then running progressions.
- Return to running by alternating walking and running, on flat ground first, changing one variable at a time. Volume or intensity or surface, never two at once, and never more than roughly ten percent at a time.
- Re-score VISA-A at weeks 8 and 12.
- Expect to keep loading the calf twice a week indefinitely. This is the part people drop, and it is why tendinopathy recurs.

The two most common failure modes, so you can avoid them: stopping at week six because it is not fixed, and never adding enough load to create a stimulus. Both feel like doing the program. Neither is.

Final thoughts

Almost everything about this condition rewards patience and punishes urgency. The tendon that took a year to get this way will not turn around in a fortnight, and the interventions that promise it will are the ones I would question hardest.

What the evidence supports is unglamorous and it is reliable: work out which type you have, load the calf heavily and slowly a few times a week, keep moving within a sensible pain rule, measure yourself honestly every four weeks, and give it three months before you judge it.

Do that and most Achilles tendons come back. Not all, which is why Chapter 12 exists and why an examination beats a downloaded protocol. But most.

References

References 1 to 8 verified against live PubMed records on 2026-09-01. References 9 to 14, added with the rebuilt nutrition chapter, verified against live PubMed records on 2026-09-04.

1. Alfredson H, Pietilä T, Jonsson P, Lorentzon R. Heavy-load eccentric calf muscle training for the treatment of chronic Achilles tendinosis. *American Journal of Sports Medicine*. 1998;26(3):360-6. **PMID 9617396**.
2. Robinson JM, Cook JL, Purdam C, Visentini PJ, Ross J, Maffulli N, et al; Victorian Institute Of Sport Tendon Study Group. The VISA-A questionnaire: a valid and reliable index of the clinical severity of Achilles tendinopathy. *British Journal of Sports Medicine*. 2001;35(5):335-41. **PMID 11579069**.
3. Öhberg L, Lorentzon R, Alfredson H. Eccentric training in patients with chronic Achilles tendinosis: normalised tendon structure and decreased thickness at follow up. *British Journal of Sports Medicine*. 2004;38(1):8-11. **PMID 14751936**.
4. Silbernagel KG, Thomeé R, Eriksson BI, Karlsson J. Continued sports activity, using a pain-monitoring model, during rehabilitation in patients with Achilles tendinopathy: a randomized controlled study. *American Journal of Sports Medicine*. 2007;35(6):897-906. **PMID 17307888**.
5. Sode J, Obel N, Hallas J, Lassen A. Use of fluoroquinolone and risk of Achilles tendon rupture: a population-based cohort study. *European Journal of Clinical Pharmacology*. 2007;63(5):499-503. **PMID 17334751**.
6. Jonsson P, Alfredson H, Sunding K, Fahlström M, Cook J. New regimen for eccentric calf-muscle training in patients with chronic insertional Achilles tendinopathy: results of a pilot study. *British Journal of Sports Medicine*. 2008;42(9):746-9. **PMID 18184750**.
7. Beyer R, Kongsgaard M, Hougs Kjær B, Øhlenschläger T, Kjær M, Magnusson SP. Heavy slow resistance versus eccentric training as treatment for Achilles tendinopathy: a randomized controlled trial. *American Journal of Sports Medicine*. 2015;43(7):1704-11. **PMID 26018970**.
8. Insertional and mid-substance Achilles tendinopathies: eccentric training is not for everyone, updated evidence of non-surgical management. *Journal of Manual and Manipulative Therapy*. 2018. **PMID 30042626**.
9. Balius R, Alvarez G, Baro F, et al. A 3-Arm Randomized Trial for Achilles Tendinopathy: Eccentric Training, Eccentric Training Plus a Dietary Supplement Containing Mucopolysaccharides, or Passive Stretching Plus a Dietary Supplement Containing Mucopolysaccharides. *Current Therapeutic Research, Clinical and Experimental*. 2016;78:1-7. **PMID 28053674**. (Two authors are employed by the ingredient manufacturer, disclosed in the text.)
10. Shaw G, Lee-Barthel A, Ross ML, Wang B, Baar K. Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. *American Journal of Clinical Nutrition*. 2017;105(1):136-43. **PMID 27852613**.
11. Lee J, Tang JCY, Dutton J, et al. The Collagen Synthesis Response to an Acute Bout of Resistance Exercise Is Greater when Ingesting 30 g Hydrolyzed Collagen Compared with 15 g and 0 g in Resistance-Trained Young Men. *Journal of Nutrition*. 2024;154(7):2076-86. **PMID 38007183**.
12. Balshaw TG, Funnell MP, McDermott EJ, et al. The Effect of Specific Bioactive Collagen Peptides on Tendon Remodeling during 15 wk of Lower Body Resistance Training. *Medicine and Science in Sports and Exercise*. 2023;55(11):2083-95. **PMID 37436929**.
13. Bischof K, Moitzi AM, Stafilidis S, König D. Impact of Collagen Peptide Supplementation in Combination with Long-Term Physical Training on Strength, Musculotendinous Remodeling, Functional Recovery, and Body Composition in Healthy Adults: A Systematic Review with Meta-analysis. *Sports Medicine*. 2024;54(11):2865-88. **PMID 39060741**.

14. Morton RW, Murphy KT, McKellar SR, et al. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. *British Journal of Sports Medicine*. 2018;52(6):376-84. **PMID 28698222**.

Working with Root Health

If your Achilles has not responded to a loading program, or you are not confident you are loading it correctly or at the right stage, I offer a free Root Discovery Session at Root Health to talk through where your tendon actually is and what a reasonable next step looks like. You can schedule one at rootcauseunlocked.com/discovery, or reach the office at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the free Root Cause Unlocked tracker has a mode built for musculoskeletal recovery, at rootcauseunlocked.com/tracker. Eight weeks of it turns “I think it might be helping” into something you and I can both read.

This guide is educational and does not constitute medical advice, diagnosis, or treatment. It does not create a doctor-patient relationship. Seek examination for any tendon problem, and seek urgent care for the red flags in Chapter 12.